

Research and Software Portfolio

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Main Repository: github.com/ayushpriyadarshi-18/geant4-detector-simulation-pipeline

1. Research Profile

I am an Integrated BS-MS Physics student at the Indian Institute of Technology Roorkee with research experience in computational detector physics, Geant4-based Monte Carlo simulations, ROOT-based analysis, scientific software development, and detector instrumentation.

My current work focuses on automated Geant4-Python workflows for scintillation detector response simulations. The workflow was developed to reduce repetitive manual work in detector-response studies by connecting simulation setup, macro generation, execution, ROOT validation, spectral analysis, count extraction, plotting, and tabulated output generation into a reproducible pipeline.

I am interested in PhD research involving detector simulations, radiation detection, nuclear and particle instrumentation, scientific software, computational physics, and high-performance simulation workflows.

2. Main Research Project

A Python-Assisted Geant4 Workflow for Automated Scintillation Detector Response Simulations

This project developed a Python-assisted Geant4 workflow for automated scintillation detector response simulations. The work combines a configurable Geant4 C++ simulation backend with Python automation scripts for campaign-level simulation and analysis.

The pipeline supports:

- Geant4 macro generation
- Simulation campaign execution
- ROOT-file validation
- Deposited-energy extraction
- Spectrum generation
- Peak-count extraction
- Total non-zero count extraction
- CSV/table generation
- Geometry and material comparison
- Public reference output generation

The workflow was developed as part of my undergraduate thesis and later consolidated into a public research software repository.

Repository: github.com/ayushpriyadarshi-18/geant4-detector-simulation-pipeline

Software page: ayushpriyadarshi-18.github.io/software

3. Simulation and Analysis Workflow

The project connects the normally separate steps of a Geant4 detector-response study into a single reproducible workflow.

1. Define detector geometry, material, source, and event statistics.
2. Generate Geant4 macro files using Python.
3. Execute Geant4 simulations using the configured C++ backend.
4. Validate ROOT outputs and check simulation completion.
5. Extract deposited-energy quantities from ROOT trees.
6. Generate gamma-ray spectra.

7. Extract counts in photopeak and sum-peak windows.
8. Merge outputs into final CSV tables.
9. Store spectra, logs, macros, ROOT files, and tables as traceable outputs.

This structure makes it easier to repeat large simulation campaigns, compare geometries and materials, and preserve analysis consistency across runs.

4. Detector Geometries

The workflow was tested on multiple scintillation detector geometries.

Solid Cylindrical Detector

A conventional cylindrical scintillation detector geometry was implemented for baseline detector-response studies. This geometry is useful for studying size dependence, material dependence, and comparison with standard gamma-ray detector configurations.

Hollow Annular Detector

A hollow cylindrical detector geometry was implemented to study higher solid-angle coverage around a central source region. This geometry is relevant for coincidence and sum-peak studies where geometric efficiency is important.

Near- 4π Soccer-Ball Detector

A custom near- 4π modular detector geometry was implemented using pentagonal and hexagonal detector elements arranged around a central cavity. This geometry was designed to explore high-coverage scintillation detection and sum-peak response in compact detector assemblies.

5. Detector Materials and Sources

The workflow supports detector-response comparison across multiple scintillator materials, including NaI, LaBr₃, CsI, BGO, GGAG, and PbWO₄.

The simulation campaigns included monoenergetic gamma-ray sources and radioactive-decay cases. Public reference demonstrations were prepared using Cs-137 simulations across multiple detector geometries.

The analysis extracts deposited-energy spectra and peak-window counts, including photopeak counts, sum-peak counts, and total non-zero energy-deposition counts.

6. ROOT-Based Analysis

The analysis stage uses ROOT output files produced by the Geant4 backend. Python scripts validate the ROOT files and extract deposited-energy quantities from the event tree.

The extracted quantities are used to generate:

- Deposited-energy spectra
- Log-scale spectrum plots
- Peak-window count tables
- Total non-zero count summaries
- Combined CSV result tables

This made the workflow useful not only for running simulations, but also for automatically producing analysis-ready outputs.

7. High-Performance Computing Validation

The workflow was tested on the PARAMGanga high-performance computing facility at IIT Roorkee. High-statistics simulations, including 10^8 -event Cs-137 runs, were performed to test whether the same Geant4-Python workflow could scale beyond small local reference simulations.

The HPC validation demonstrated that the pipeline structure could support both lightweight local tests and larger simulation campaigns.

8. Research Software Output

Geant4 Detector Simulation Pipeline

Role: Developer and Maintainer

Repository: github.com/ayushpriyadarshi-18/geant4-detector-simulation-pipeline

The public repository contains:

- Geant4 C++ source code
- Python automation scripts
- Example macro files
- ROOT output validation tools
- Spectrum-generation scripts
- Count-extraction scripts
- Reference spectra
- CSV result tables
- Logs and visualisation files

The repository serves as both a research output and a reproducibility package for the thesis work.

9. Conference Presentation

A first-author conference paper based on this work has been accepted for poster presentation at IEEE Nuclear Science Symposium 2026.

A. Priyadarshi, G. A. Kumar, K. Madhan, and V. Ranga, “A Python-based Tool for Automated Geant4 Simulations of Scintillation Detectors,” accepted for poster presentation at IEEE Nuclear Science Symposium, 2026.

Status: Accepted for poster presentation.

10. Related Instrumentation and Systems Projects

Radio Telescope Control and Embedded Linux Support

I contributed to Raspberry Pi-based embedded Linux and hardware-control support for a student-led radio telescope project under the Physics and Astronomy Club, IIT Roorkee. This work involved custom Raspberry Pi image preparation, boot configuration modifications, networking over a power/data port, support for an initially unsupported Wi-Fi module, and hardware-control integration.

Raspberry Pi Home Server and Network Gateway

I designed and maintain a Raspberry Pi-based home server for self-hosted services, home automation, file sharing, reverse proxy services, DNS filtering, and network gateway functionality. This project strengthened my practical Linux, networking, Docker, and systems-administration experience.

WiFi IP Mouse

I built a Raspberry Pi-based WiFi IP mouse system during the Syntax Error Hackathon organised by SDS Labs, IIT Roorkee. The project won First Prize and involved network-based device control under hackathon time constraints.

11. Technical Strengths

- Geant4 detector simulation
- Python automation
- ROOT-based analysis
- C++ simulation backend development
- Scientific software workflows
- Linux-based development
- Git-based version control
- HPC simulation execution
- Detector-response modelling
- Gamma spectroscopy analysis
- Raspberry Pi and embedded Linux systems
- Scientific instrumentation and hardware-control support

12. Research Direction and Application Areas

For PhD research, I am interested in applying and extending my experience in computational detector physics, simulation automation, scientific software, and scientific instrumentation.

The skills developed through my Geant4–Python detector simulation work are broadly applicable to multiple areas of physics, including:

- Nuclear physics experiments
- Particle detector simulation and instrumentation
- Astroparticle and cosmic-ray detection
- Radiation detection and gamma spectroscopy
- Reactor, neutron, and radiation-transport studies
- Accelerator-based experimental physics
- Space-radiation and shielding studies
- Scientific software for experimental physics
- High-performance Monte Carlo simulation workflows

My strongest current direction is toward groups working on detector-response modelling, Geant4 simulations, radiation detection, nuclear or particle instrumentation, and computational tools for experimental physics.